Jiale Zhang

Ph.D. Candidate in Computer Science

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I study AI systems that participate in scientific inquiry: organizing evidence, coordinating specialized agents, and supporting transparent, iterative research workflows across domains.

AI FOR SCIENCE	MULTI-AGENT SYSTEMS	RAG & GENERATIVE RETRIEVAL

EDUCATION

PRESENT	**Ph.D. Candidate in Computer Science** Fudan University · Shanghai, China
2023–2026	**Master's in Computer Technology** University of Chinese Academy of Sciences · Shenyang Institute of Computing Technology

PUBLICATIONS

01 **SlimRAG: Retrieval without Graphs via Entity-Aware Context Selection.** Jiale Zhang, Jiaxiang Chen, Zhucong Li, Jie Ding, Kui Zhao, Zenglin Xu, Xin Pang, and Yinghui Xu. *arXiv preprint*, 2025.
Introduces entity-aware context selection that removes graph construction while improving retrieval precision and reducing indexing and context overhead.
Paper · PDF · Code · DOI

02 **ChunkGraph: Relationship-Driven Retrieval Through Progressive Complete Graphs.** Jiale Zhang, Kui Zhao, Hao Zhang, Fuzhe Zhang, and Xu Liu. *IEEE ICCC*, 2025.
Models relationships directly between document chunks and progressively discovers and prunes retrieval paths without entity extraction.
DOI

03 **AI2Agent: An End-to-End Framework for Deploying AI Projects as Autonomous Agents.** Jiaxiang Chen, Jingwei Shi, Lei Gan, Jiale Zhang, Qingyu Zhang, Dongqian Zhang, Xin Pang, Zhucong Li, and Yinghui Xu. *ACL System Demonstrations*, 2025.
Turns AI projects into autonomous deployment agents through guideline-driven execution, adaptive debugging, and reusable deployment cases.
Paper · PDF · DOI

SELECTED RESEARCH SYSTEMS

Scientific inquiry agents
Developing general-purpose agentic workflows that help formulate questions, assemble evidence, test intermediate claims, and carry investigations across iterative research cycles.

Coordinated agent teams
Designing role specialization, task routing, shared workspaces, observability, and recovery mechanisms for long-running multi-agent systems.

Evidence-centered knowledge systems
Building retrieval and generation pipelines with query decomposition, attributable context selection, provenance, and controlled abstention.

RESEARCH SOFTWARE

SlimRAG reference implementation
Public implementation of entity-aware retrieval without graph traversal.

Agent-ready full-stack scaffold
A production-oriented FastAPI and Vue foundation with an agent-facing project map, reproducible setup, and continuous integration.

Reproducible Python research scaffold
A compact uv-based repository template with deterministic environments, testing, linting, and continuous integration.